

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		550 - 555		1		1	1	
		(ii)		[Galaxy] D (1) [The absorption lines] {have the <u>greatest</u> red shift / <u>longest</u> wavelength / moved <u>most</u> to the right / red} (1)		2		2		
		(iii)		[The universe is] expanding / started with the <u>Big Bang</u> Accept started at a single point Don't accept it's moving away		1		1		
	(b)			[Only] helium is present or there is no iron present (1) [Only] line at 587 [units] or there is no line at 527 [units] (1)			2	2		
				Question 4 total	0	4	2	6	1	0